Representing Names

Name: _____ Date: _____

Two teams each wrote a `Student` record for a university information system.

Team A: first and last

```
record Student(String first, String last, String id) {
  String lastFirst() {
    return last + ", " + first;
  }
  String initialLast() {
    return first.substring(0, 1) + ". " + last;
  }
  String email(String domain) {
    return first.substring(0, 1).toLowerCase()
      + last.toLowerCase() + "@" + domain;
  }
}
Student ava = new Student("Ava", "Chen", "A12345678");
String s1 = ava.lastFirst();   // "Chen, Ava"
String s2 = ava.initialLast(); // "A. Chen"
String e1 = ava.email("ucsd.edu");
String e2 = ava.email("health.ucsd.edu");
// "achen@ucsd.edu", "achen@health.ucsd.edu"
```

Team B: one name

```
record Student(String name, String id) {
  String lastFirst() {
    int space = name.indexOf(" ");
    return name.substring(space + 1, name.length())
      + ", " + name.substring(0, space);
  }
  String initialLast() {
    int space = name.indexOf(" ");
    return name.substring(0, 1) + ". "
      + name.substring(space + 1, name.length());
  }
}
Student ava = new Student("Ava Chen", "A12345678");
String s1 = ava.lastFirst();   // "Chen, Ava"
String s2 = ava.initialLast(); // "A. Chen"
```

Q1. Fill in what Team B's methods produce for each `name`. Then say what you would put in Team A's `first` and `last` for the same person.

B's `name`	B: `lastFirst()`	B: `initialLast()`	A: `first` / `last`?
`"Mary Jane Watson"`			
`"Ludwig van Beethoven"`			
`"Nguyen Van An"` (family name first)			
`"Simon"`			

Q2. Team A can write `String oneName() { return first + " " + last; }` to get a name like Team B expects. Could Team B write a method that returns a Team A `Student`? Which record holds more information?

Q3. Team B wants `email(String domain)` too. Write it, so that `new Student("Ava Chen", "A12345678").email("ucsd.edu")` is `"achen@ucsd.edu"`. What does yours return for `"Simon"`?

Q4. Use Team A's `email("ucsd.edu")`. What is wrong with each address?

`first` / `last`	`email("ucsd.edu")`	what's wrong?
`"Ludwig"` / `"van Beethoven"`		
`"José"` / `"Núñez"`		
`"Jane"` / `"Smith"` and `"John"` / `"Smith"`		

Q5. Why is `domain` a parameter and not a field of `Student`? Describe a student who needs both `ucsd.edu` and `health.ucsd.edu`.

Q6. What do you think needs to be true of the String returned from the `email` method?

Q7. Nobody would guess `health.ucsd.edu` from the roster; someone who works there had to say so. Before building `email`, who else would you ask, and what might they tell you?

Q8. You are working with an AI agent. How would you allocate responsibility between you and the agent for each of these tasks?

	← more my job … more the agent's job →
(a) Writing code to turn names with accented characters and spaces into just a-z, A-Z, and numbers for email addresses	
(b) Choosing which fields to use to represent students' names	
(c) Figuring out which domains are used for which university systems	
(d) Deciding whether a legal name or a chosen name gets shown to a professor in a course roster	
(e) Given a design, writing code to figure out how to create a `Student` record for each student in an existing system and reporting any errors or unrepresentable cases.	

Q9. There are other issues, too, like chosen vs. legal name and other constraints from other languages and cultures. Given all this, discuss what might be needed for a more comprehensive design of names in a student system, in terms of a Java `record` with fields and methods.